Transformation

Motivation

- Let (X_1, X_2) be a random vector. Suppose we know the joint distribution of (X_1, X_2) and we seek the distribution of a transformation of (X_1, X_2), say, $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$.

Discrete Random Variables

- Discrete (Easy): Let $p_{X_1,X_2}(x_1, x_2)$ be the joint pmf of X_1 and X_2 with support $\mathcal{S}$. Let $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ define a one-to-one transformation that maps $\mathcal{S}$ to $\mathcal{T}$. The joint pmf of $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$ is given by

$$p_{Y_1,Y_2}(y_1, y_2) = \begin{cases} p_{X_1,X_2}[w_1(y_1, y_2), w_2(y_1, y_2)] & (y_1, y_2) \in \mathcal{T} \\ 0 & \text{elsewhere,} \end{cases}$$

where $x_1 = w_1(y_1, y_2)$, $x_2 = w_2(y_1, y_2)$ is the single-valued inverse of $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$.

Continuous Random Variables

- cdf technique
- A simple example: Let the joint pdf of X and Y be

$$f_{X,Y}(x,y) = \begin{cases} 1 & 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the pdf of $Z = X + Y$.
$F_Z(z) = \mathbb{P}(X + Y \leq z) = \int\int_{x+y \leq z} f_{X,Y}(x,y)dxdy$. Then

$$F_Z(z) = \begin{cases} 0 & z < 0 \\ \int_0^z \int_0^{z-x} dydx = \frac{z^2}{2} & 0 \leq z < 1 \\ 1 - \int_{z-1}^1 \int_{z-x}^1 dydx = 1 - \frac{(2-z)^2}{2} & 1 \leq z < 2 \\ 1 & 2 \leq z. \end{cases}$$

Since $f_Z(z) = F_Z'(z)$, the pdf of Z is

$$f_Z(z) = \begin{cases} z & 0 < z < 1 \\ 2 - z & 1 \leq z < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

General Cases

- Consider more general case: Let (X_1, X_2) have pdf $f_{X_1,X_2}(x_1, x_2)$. Suppose $Y_1 = u_1(X_1, X_2)$, $Y_2 = u_2(X_1, X_2)$, where $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ define a one-to-one transformation from $\mathcal{S} \in R^2$ onto $\mathcal{T} \in R^2$. Write the inverse of transformation as $x_1 = w_1(y_1, y_2)$ and $x_2 = w_2(y_1, y_2)$. Define the **Jacobian** as

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$$

General Cases

- Let A be a subset of $\mathcal{S}$ and let B denote the mapping of A under the one-to-one transformation. Because the transformation is one-to-one, the event $\{(X_1, X_2) \in A\}$ and $\{(Y_1, Y_2) \in B\}$ are equivalent. Hence,

$$P[(Y_1, Y_2) \in B] = P[(X_1, X_2) \in A] = \int\int_A f_{X_1,X_2}(x_1, x_2) dx_1 dx_2$$

$$= \int\int_B f_{X_1,X_2}(w_1(y_1, y_2), w_2(y_1, y_2))|J| dy_1 dy_2$$

[**check your multivariate analysis book**] So,

$$f_{Y_1,Y_2}(y_1, y_2) = \begin{cases} f_{X_1,X_2}(w_1(y_1, y_2), w_2(y_1, y_2))|J| & (y_1, y_2) \in \mathcal{T} \\ 0 & \text{elsewhere} \end{cases}$$

An Example

- Let X_1 and X_2 have the joint pdf

$$f_{X_1,X_2} = \begin{cases} 10x_1x_2^2 & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Suppose $Y_1 = X_1/X_2$ and $Y_2 = X_2$. Find the marginal pdf of Y_1 and Y_2.

1. inverse transformation: $x_1 = y_1y_2$ and $x_2 = y_2$.
2. Jacobian: $J = \begin{vmatrix} y_2 & y_1 \\ 0 & 1 \end{vmatrix} = y_2$.
3. support: $0 < y_1y_2$, $y_1y_2 < y_2$, $y_2 < 1$. This is equivalent to $0 < y_1 < 1$, $0 < y_2 < 1$.
4. joint pdf: $f(y_1, y_2) = 10y_1y_2y_2^2|y_2| = 10y_1y_2^4$, $0 < y_1 < 1$, $0 < y_2 < 1$.
5. marginal pdfs: $f_{Y_1}(y_1) = \int_0^1 10y_1y_2^4 dy_2 = 2y_1$, $0 < y_1 < 1$;
   $f_{Y_2}(y_2) = \int_0^1 10y_1y_2^4 dy_1 = 5y_2^4$, $0 < y_2 < 1$.